

# Ethereum Staking Behavior Analysis from 02.09.2026

On-chain analysis of 500 Ethereum staking addresses (Lido, 90-day window), exploring what drives long-term staker engagement beyond simple deposit size.

**Pipeline:** Dune Analytics (SQL) → PostgreSQL (staging table + SQL views) → Power BI (5-page dashboard)

## Key findings:

- Staking frequency — not stake size — is the strongest predictor of engagement duration: single-stake addresses stay active ~3 days on average, while frequent stakers (20+ stakes) remain active 100+ days, a 40x difference
- Volume is highly concentrated: the top stake-size tier (1,000+ ETH) holds just 13% of addresses but ~4x the volume of the mid tier
- 75% of stakers show one-off engagement (<1 week), consistent with a long-tail retention pattern seen across consumer products

**Dashboards:** Whale Concentration · Staker Churn · Behavior Matrix · Staking Cohorts · Engagement Distribution

## Full SQL queries and build process:

### Dune Analytics (SQL)

Event details

2026-09-02 12:02

8584798

01M1..1QF9

Success

Execution details

|                 |         |          |              |          |
|-----------------|---------|----------|--------------|----------|
| Record type     | Credits | Product  | User name    | Duration |
| Query Execution | 0.9951  | Data Hub | hui_satoschi | 20s      |
| CPU seconds     |         |          |              |          |
| 19s             |         |          |              |          |

SQL text

```
SELECT
  "from" AS staker,
  MIN(block_time) AS first_stake,
  MAX(block_time) AS last_stake,
  COUNT(*) AS total_stakes,
  SUM(value) / 1e18 AS total_staked_eth,
  ROUND((SUM(value) / 1e18) / COUNT(*), 4) AS avg_stake_size_eth
FROM ethereum.transactions
WHERE "to" = 0xae7ab96520de3a18e5e111b5eaaab095312d7fe84 -- Lido stETH contract
  AND block_time >= NOW() - INTERVAL '90' DAY
  AND value > 0
GROUP BY "from"
HAVING COUNT(*) >= 1
ORDER BY total_staked_eth DESC
LIMIT 500
```

Execution metadata

|                       |                  |                    |                            |                                |
|-----------------------|------------------|--------------------|----------------------------|--------------------------------|
| Cell count            | Column count     | Execution attempts | Execution duration seconds | Execution pending milliseconds |
| 3000                  | 6                | 1                  | 20                         | 2                              |
| Planning time seconds | Result set bytes | Row count          | Performance                |                                |
| 0.01581               | 41000            | 500                | Medium                     |                                |

## PostgreSQL (staging table + SQL views)

### Creating a working table “staker\_summary” and view “v\_staker\_analysis”

| Показать SQL |                                             | Введите SQL выражение чтобы отфильтровать результаты |                         |              |                  |                    |
|--------------|---------------------------------------------|------------------------------------------------------|-------------------------|--------------|------------------|--------------------|
| Таблица      | staker                                      | first_stake                                          | last_stake              | total_stakes | total_staked_eth | avg_stake_size_eth |
| 1            | 0xf34d97c680803358a468e3c56337defaf85ca6da  | 2026-07-21 07:19:00.000                              | 2026-07-23 15:20:00.000 | 26           | 127 141,2287     | 4 890,0473         |
| 2            | 0x4f87de7d21aef48090958f7342e1f69dff790545  | 2026-06-12 21:10:00.000                              | 2026-06-14 22:57:00.000 | 20           | 83 143,7187      | 4 157,1859         |
| 3            | 0xfb78aa8f38843629e89951d9db6fdc398d75e0a3  | 2026-06-05 21:31:00.000                              | 2026-06-05 21:31:00.000 | 1            | 29 000           | 29 000             |
| 4            | 0x4c49d4bd6a571827b4a556a0e1e3071da6231b9d  | 2026-06-04 12:17:00.000                              | 2026-09-02 11:46:00.000 | 1 319        | 26 747,26755     | 20,2784            |
| 5            | 0x9c7f3400109f04e8044989bbb7d483dc4587053a  | 2026-06-07 14:41:00.000                              | 2026-08-25 06:27:00.000 | 16           | 17 397           | 1 087,3125         |
| 6            | 0x761cc954e2d968ac105f14794b25927a3b6216f4  | 2026-07-03 13:39:00.000                              | 2026-07-03 13:48:00.000 | 3            | 17 000           | 5 666,6667         |
| 7            | 0x43fc188f003e444e9e538189fc675acdfeb8f5d12 | 2026-07-24 13:01:00.000                              | 2026-08-05 14:30:00.000 | 2            | 14 000           | 7 000              |
| 8            | 0x176f3dab24a159341c0509bb36b833e7fdd0a132  | 2026-07-07 13:23:00.000                              | 2026-07-07 13:23:00.000 | 1            | 13 000           | 13 000             |
| 9            | 0xb99a2c4c1c4f1fc27150681b740396f6ce1cbcf5  | 2026-08-24 07:09:00.000                              | 2026-08-24 07:09:00.000 | 1            | 12 498           | 12 498             |
| 10           | 0xed0c6079229e2d407672a117c22b62064f4a4312  | 2026-08-14 14:04:00.000                              | 2026-08-24 07:25:00.000 | 2            | 10 200           | 5 100              |

### 1. Whale Concentration — distribution by stake size

What it examines: the proportion of addresses holding a specific share of the total ETH supply—revealing wealth inequality or concentration.

```
postgres crypto_analytics staker_summary *<postgres> Console X
CREATE VIEW crypto_analytics.v_whale_concentration AS
SELECT
  CASE
    WHEN total_staked_eth < 1 THEN '1) < 1 ETH'
    WHEN total_staked_eth < 10 THEN '2) 1-10 ETH'
    WHEN total_staked_eth < 100 THEN '3) 10-100 ETH'
    WHEN total_staked_eth < 1000 THEN '4) 100-1,000 ETH'
    ELSE '5) 1,000+ ETH'
  END AS stake_tier,
  COUNT(*) AS num_stakers,
  ROUND(SUM(total_staked_eth), 2) AS tier_volume_eth,
  ROUND(100.0 * SUM(total_staked_eth) / SUM(SUM(total_staked_eth)) OVER (), 1) AS pct_of_total_volume
FROM crypto_analytics.staker_summary
GROUP BY stake_tier
ORDER BY stake_tier;
```

### 2. Churn / Retention — active vs. inactive stakers

What it measures: the percentage of addresses that have not made new stakes in the last 30 days (churn)—a direct analogue to "player churn" in iGaming.

```
postgres crypto_analytics staker_summary *<postgres> Console X
CREATE VIEW crypto_analytics.v_staker_churn as
SELECT
  CASE
    WHEN last_stake < NOW() - INTERVAL '30 day' THEN '1) Churned (30+ days inactive)'
    ELSE '2) Active (staked within 30 days)'
  END AS staker_status,
  COUNT(*) AS num_stakers,
  ROUND(AVG(total_staked_eth), 2) AS avg_stake_eth,
  ROUND(SUM(total_staked_eth), 2) AS total_volume_eth
FROM crypto_analytics.staker_summary
GROUP BY staker_status;
```

### 3. Frequency vs. Stake Size — behavioral segmentation (bubble chart)

What it analyzes: using pre-defined segments from `v\_staker\_analysis`, it determines how many addresses fall into each "frequency × stake size" combination (DCA stakers vs. one-shot whales).

```
postgres crypto_analytics staker_summary *<postgres> Console X
CREATE VIEW crypto_analytics.v_staker_behavior_matrix as
SELECT
    frequency_segment,
    stake_size_segment,
    COUNT(*) AS num_stakers,
    ROUND(AVG(days_active), 1) AS avg_days_active
FROM crypto_analytics.v_staker_analysis
GROUP BY frequency_segment, stake_size_segment
ORDER BY frequency_segment, stake_size_segment;
```

### 4. Staking Cohorts — cohort size and activity based on the week of the initial stake

What it analyzes: the weekly inflow of new stakers and their average "longevity" (days\_active)—a retention trend over time.

```
postgres crypto_analytics staker_summary *<postgres> Console X
CREATE VIEW crypto_analytics.v_staking_cohorts as
SELECT
    first_stake_week,
    COUNT(*) AS stakers_in_cohort,
    ROUND(AVG(days_active), 1) AS avg_days_active,
    ROUND(AVG(stakes_per_active_day), 3) AS avg_stakes_per_day
FROM crypto_analytics.v_staker_analysis
GROUP BY first_stake_week
ORDER BY first_stake_week;
```

### 5. Days Active Distribution — distribution of the "engagement period"

What it analyzes: a histogram showing the average number of days an address remains an "active" staker—revealing a bimodal pattern (short-term speculators vs. long-term holders).

```
postgres crypto_analytics staker_summary *<postgres> Console X
CREATE VIEW crypto_analytics.v_engagement_distribution as
SELECT
    CASE
        WHEN days_active = 0 THEN '1) Single day (0)'
        WHEN days_active < 7 THEN '2) < 1 week'
        WHEN days_active < 30 THEN '3) 1-4 weeks'
        WHEN days_active < 90 THEN '4) 1-3 months'
        ELSE '5) 3+ months'
    END AS engagement_bucket,
    COUNT(*) AS num_stakers,
    ROUND(AVG(total_stakes), 1) AS avg_total_stakes
FROM crypto_analytics.v_staker_analysis
GROUP BY engagement_bucket
ORDER BY engagement_bucket;
```